$\sqrt{1+y^2}$) is a mapping of G onto E (the interval $[1, \sqrt{2}]$), that is, the mapping g. Finally, the function h (the function $\sqrt{1+\cos^2 x}$ defined over the interval $[0, \pi]$) is a mapping of D onto E, that is, the mapping h.

The mapping h is a result of the $\boxed{\text{consecutive}}$ mappings f and g, and is said to be the $\boxed{\text{composition of mappings}}$; the following notation is used $\boxed{h = g \circ f}$ (the right-hand side of the equation should be read from right to left: the mapping f is used first and then the mapping g).

Reader: Obviously, for a composite function one can also draw a diagram shown in Fig. 22.

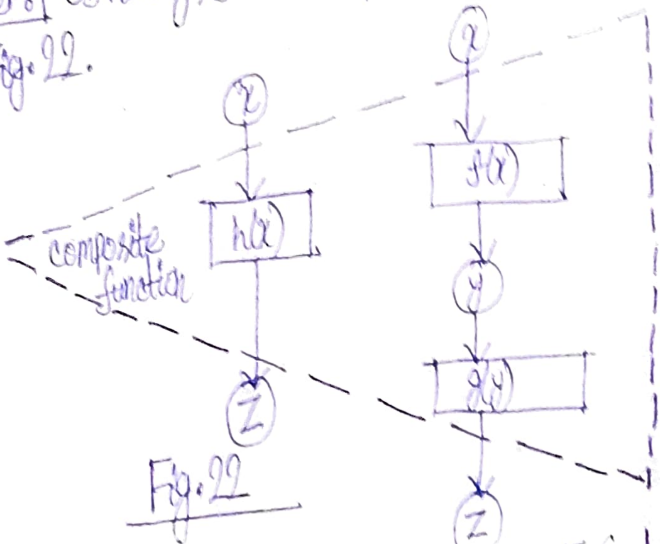

Fig. 22

Author: I have no objections. Although I feel that we better proceed from the concept of mapping of one set onto another, as in Fig. 21

Reader: Probably, certain "difficulties" may arise because the range of f is at the same time the domain of g?

Author: In any case, this observation must always be kept in mind. One should not forget that the natural domain of a composite function $g[f(x)]$ is a portion (subset) of the natural domain of $f(x)$ for which the values of f belong to the natural domain of g. This aspect was unimportant in the example concerning $g[f(x)] = \sqrt{1+\cos^2 x}$ because all the values of f (even if $\cos x$ is defined on the whole real line) fall into the natural domain of $g(y) = \sqrt{1+y^2}$.

I can give you, however, a different example: $h(x) = \sqrt{\sqrt{x-1} - 2}$,
$f(x) = \sqrt{x-1}$, $g(y) = \sqrt{y-2}$